SPATIAL TAXONOMY · PROOF OF CONCEPT

Building Width Calculator

Furniture → Zone → Width. A backwards-running method for sizing office buildings from tenant fit-out, driving the `tool-buildingwidth` engine.

AGENDA — FIVE SLIDES

The Backwards Method

Most office buildings start with a width — sixty feet for daylight — then fit furniture into what remains. Woodfine inverts the sequence: start with the tenant's desk, workstation, exam table or bench; the zones follow from the furniture, and the building width is whatever the arrangement requires. The cross-section has three zones, with Zone 1 (Habitat) and Zone 2 (Magazine) mirrored on each side of a single centreline Zone 3 (Corridor).

Facade	Frontage	–
Zone 1	Habitat — desks	6 m
Zone 2	Magazine — storage	3 m
Zone 3	Corridor	3 m
Zone 2	Magazine — storage	3 m
Zone 1	Habitat — desks	6 m
Facade	Frontage	–
TOTAL	Professional Office	**21.0 m** 68'11"

Zone 1 — Habitat

Desks, workstations, exam tables, benches. Six metres from the façade is the working baseline — the European Lighting Standard threshold for daylight at a workstation; some use types widen it (Medical 7.28 m for exam-table depth) and Academic compresses it because seating faces forward, not toward the façade.

Zone 2 — Magazine

Storage, bookshelves, filing, staff rooms, server rooms. The slack variable: widen it to balance the floor plate so it doesn't end up as a long, narrow rectangle.

Zone 3 — Corridor

A single centreline corridor, not mirrored. Width set by egress and accessibility — overdesigned by 20% to promote well-being and absorb the "in and out" traffic of smaller tenants.

The 0.7 m German Circulation Law addition. Perpendicular desks at 6.0 m Habitat fit only two in series; 0.7 m unlocks the third with clearance. Token: `bim.circulation-addition.perpendicular-desk = 0.7 m`.

Zone Depth by Use Type

Each segment's width is the actual zone depth in metres, scaled against the widest total. Building width = 2×(Habitat + Magazine) + Corridor — the corridor is centreline, single, not mirrored. The widest tenant (Business at its open-storage maximum) is roughly 17.6 m wider than the narrowest (Private Office); a mixed-tenant floor must reconcile this within a single plate width.

The Building Width Calculator

Zone 1 + Zone 2 | Zone 3 | Zone 2 + Zone 1

Building Width = 2 × (Habitat + Magazine) + Corridor

USE TYPE	Z1 HABITAT	Z2 MAGAZINE	Z3 CORRIDOR	HALF (Z1+Z2)	TOTAL WIDTH
Private Office	5.99 m / 19'8"	1.37 m / 4'6"	3 m / 9'10"	7.37 m	**17.73 m** / 58'2"
Academic	4.70 m / 15'5"	3 m / 9'10"	3 m / 9'10"	7.70 m	**18.40 m** / 60'4"
Professional Office	6 m / 19'8"	3 m / 9'10"	3 m / 9'10"	9.00 m	**21.00 m** / 68'11"
Laboratory	6.78 m / 22'3"	4.80 m / 15'9"	3.05 m / 10'0"	11.58 m	**26.21 m** / 86'0"
Medical	7.28 m / 23'11"	4.88 m / 16'0"	2.89 m / 9'6"	12.16 m	**27.21 m** / 89'3"
Business	5.51 m / 18'1"	9.26 m / 30'5"	2.75 m / 9'0"	14.77 m	**32.29 m** / 105'11"
Civic	6 m / 19'8"	7.23 m / 23'9"	3.60 m / 11'10"	13.23 m	**30.06 m** / 98'7"

WIDEST BUILDING WIDTH

32.29 m

105'11" · Business

NARROWEST BUILDING WIDTH

17.73 m

58'2" · Private Office

SPREAD — WIDEST MINUS NARROWEST

14.56 m

47.8 ft

The ±5 ft precision question. Tiles must compose to a single floor-plate width across all use types on a floor. `tool-buildingwidth` resolves this by computing each floor's tenant-mix maximum width at design time, then redistributing Zone 2 or constraining the mix.

SECTION 04

The Floor Plate

Tiles are the assembly units — Small Tile = 2,700 SF, Large Tile = 4,900 SF — and they compose to fill the plate exactly. Each Tile is an independent climate zone: a tenant who wants thermostat control leases a Tile. Three example compositions below.

Composition A — Corporate-heavy

Three large corporate tiles (Tile F, 4,900 SF each) plus two standard corporate tiles (Tile A, 2,700 SF each). Five independent climate zones — each tenant controls their own thermostat. Total: 20,100 SF.

Tile F 4,900 SF 1 climate zone	**Tile F** 4,900 SF 1 climate zone	**Tile F** 4,900 SF 1 climate zone	**Tile A** 2,700 SF 1 climate zone	**Tile A** 2,700 SF 1 climate zone

Tiles: 5 · Climate zones: 5 · Total: 20,100 SF (target 20,000)

Composition B — Private Office heavy

Maximum climate-zone count. Two Tile G (4,900 SF, 10 climate zones each) plus four Tile B-1 (2,700 SF, 5 climate zones each) end caps. Twenty private licensed offices and twenty more in end caps. Forty independent thermostats on one floor.

Tile B-1 2,700 SF 5 climate zones	**Tile G** 4,900 SF 10 climate zones	**Tile B-1** 2,700 SF 5 climate zones	**Tile B-1** 2,700 SF 5 climate zones	**Tile G** 4,900 SF 10 climate zones	**Tile B-1** 2,700 SF 5 climate zones

Tiles: 6 · Climate zones: 40 · Total: 20,600 SF (target 20,000)

Composition C — Mixed Professional Centre

Two Tile H (large professional, 3 climate zones each), one Tile C-2 (Medium professional pair), one Tile C-1 (Medium + Small), and two end-cap mixed tiles (E-1 / E-2). Tenants: Medical anchor + Business cluster + Private Office end caps.

Tile E-1 2,700 SF 3 climate zones	**Tile H** 4,900 SF 3 climate zones	**Tile C-2** 2,700 SF 2 climate zones	**Tile H** 4,900 SF 3 climate zones	**Tile C-1** 2,700 SF 2 climate zones	**Tile E-2** 2,100 SF 3 climate zones

Tiles: 6 · Climate zones: 16 · Total: 20,000 SF (target 20,000)

> **Tile = Climate Zone.** Composition B (Private Office heavy) yields forty independent thermostats on one floor. Source token: `bim.tile.design-principles.climate-zone-autonomy`.

What Comes Next

This HTML is the specification document for `tool-buildingwidth`, a Rust engine that reads the DTCG tokens listed at right and computes floor plate dimensions, tile compositions, and demising-wall coordinates from a tenant-mix input.

The engine outputs IFC-compatible geometry and a leasing-plan summary; this deck is what the engine will render as a human preview.

Every metre, every square foot, every percentage on these slides is read live from the DTCG token files at page load. When the tokens change — a new key plan sample, a corridor width update — the deck changes with them. The engine, when shipped, shares the same source.

Source: woodfine-design-bim DTCG token library v0.0.2

SOURCE TOKEN FILES

```
building-width-calculator.dtcg.json
floor-plate-standards.dtcg.json
professional-office-subtypes.dtcg.json
tile-system.dtcg.json
```

FUTURE CONSUMER

```
pointsav-monorepo/
  crates/tool-buildingwidth/
```